

Chapter 11 — Perioperative Care in Urologic Surgery — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **≥4 METs (metabolic equivalents; 2 flights of stairs)** → no further cardiac testing
- **Elective surgery after MI (myocardial infarction):** wait 4–6 weeks
- **Warfarin:** stop 5 days → INR (international normalised ratio) <1.5
- **Aspirin/clopidogrel:** stop 7–10 days; DES (drug-eluting stent) → delay >1 year
- **Stop smoking ≥8 weeks;** <8 weeks may ↑ risk
- **Antibiotic:** within 60 min of incision; ≤24 h; redose >4 h
- **Normothermia** 36–38°C; 1°C drop → ↑ bleeding, ↑ SSI (surgical site infection)
- **Transfuse:** rarely Hct (haematocrit) >30%; often <21%
- **TRALI** (transfusion-related acute lung injury): commonest transfusion death
- **Stress steroids:** hydrocortisone 50–100 mg IV at induction → 25–50 mg 8-hourly 24–48 h

PRESURGICAL TESTING

- **Routine tests:** never shown cost-effective; ASA (American Society of Anesthesiologists) status predicts better
- **ECG:** age >40 or cardiac history
- **Pregnancy test:** all women of childbearing age, morning of surgery
- **Chest radiograph, PT/PTT** (prothrombin/partial thromboplastin time): not routine
- **ASA:** III limits activity; IV constant threat to life; V moribund <24 h; VI brain-dead donor; E = emergency suffix
- **ASA grades physical status,** not surgical risk; independent mortality predictor
- **Goldman heaviest:** S3/JVD (jugular venous distension) 11; recent MI 10
- **Goldman risk:** 6–12 → 7%; 13–25 → 14%; ≥26 → 78%
- **Modified Cardiac Risk Index:** 6 predictors, 1 point each — ischaemic heart disease, congestive heart failure (CHF), cerebrovascular disease, high-risk surgery, insulin, creatinine >2 mg/dL

CARDIOVASCULAR EVALUATION

- **Major predictors:** MI <7 days; MI 7 days–1 month; unstable angina; decompensated heart failure (HF); significant arrhythmia; severe valve disease
- **Intermediate:** mild angina, prior MI/Q waves, compensated HF, diabetes, creatinine >2
- **Minor:** age, abnormal ECG, non-sinus rhythm, prior stroke, uncontrolled hypertension
- **Old 3–6-month post-MI dictum:** abandoned
- **DASI** (Duke Activity Status Index): $VO_2 = 0.43 \times DASI + 9.6$; $METs = VO_2 \div 3.5$
- **Surgery risk:** high = major emergency/large shifts; intermediate = intraperitoneal, laparoscopic, robotic; low = endoscopic, superficial

PULMONARY AND HEPATOBILIARY

- **FEV₁ (forced expiratory volume in 1 s)** <0.8 L or <30% predicted → high risk
- **Smokers:** 4× pulmonary morbidity; up to 10× mortality
- **Restrictive fares better than obstructive** (effective cough)
- **Jaundice visible:** bilirubin >3 mg/dL
- **Child mortality:** A 10%, B 30%, C 76–82%
- **MELD** (Model for End-Stage Liver Disease): bilirubin, creatinine, INR — objective, more accurate

SPECIAL POPULATIONS

- **Age independently predicts** morbidity and mortality

- **Delirium** may be first sign of sepsis/metabolic upset; regional anaesthesia, fewer opioids
- **BMI (body mass index) ≥45:** 8–13 years' life lost; >3 coronary risk factors → non-invasive cardiac testing
- **Pregnancy fetal risk:** preimplantation to ~15 weeks; malformation threshold 10 cGy
- **CT ~1 cGy; IVP (intravenous pyelogram) ~0.3 cGy** → ultrasound first
- **Safest trimester:** second
- **Pregnancy analgesia:** opioids yes; NSAIDs (non-steroidal anti-inflammatory drugs) no — ductus arteriosus
- **Severe malnutrition:** >20 lb (~9 kg) loss in 3 months
- **TPN (total parenteral nutrition) preop 7–10 days:** only if severely malnourished; harms well-nourished
- **Enteral preferred;** 5–20 days preop → complications ↓ 10–15%
- **Postop PN** if needs unmet by 7–10 days; commonest line complication = sepsis

PREPARATION FOR SURGERY

- **Only β-blockade** (not nitrates/calcium-channel blockers) improved cardiac outcome — Mangano atenolol 1996
- **>6 months' abstinence** = non-smoker risk
- **Glucose target:** 80–250 mg/dL; stop long-acting oral agents; pump — keep basal rate
- **Hyperthyroid:** atrial fibrillation (AF) ~20%; storm → iodine + steroids; continue propylthiouracil/methimazole + β-blocker
- **Hypothyroid:** ↑ anaesthetic/opioid sensitivity
- **HPA** (hypothalamic–pituitary–adrenal) **suppression:** >20 mg prednisone >3 weeks or cushingoid; ≤5 mg — none
- **Supplement** if suppressive dose within 1 year; minor local procedure — no stress dose
- **Warfarin half-life** 36–42 h
- **Bridge** (American College of Chest Physicians 2012): moderate and high thrombotic risk — low-molecular-weight heparin (LMWH) or IV unfractionated heparin (UFH)
- **High risk:** any mitral valve; caged-ball/tilting-disc; CHADS₂ (heart failure, hypertension, age, diabetes, stroke) 5–6; venous thromboembolism (VTE) <3 months; severe thrombophilia; stroke/transient ischaemic attack <3–6 months
- **Low risk:** bileaflet aortic, CHADS₂ 0–2, single VTE >12 months
- **Dual antiplatelet:** BMS (bare-metal stent) 6 weeks; DES 12 months
- **Delay elective surgery:** BMS ≥30 days; DES >1 year
- **Premature antiplatelet stop:** 25–50% significant MI
- **Can't wait (cancer), DES:** continue aspirin + clopidogrel
- **Mechanical bowel preparation (MBP):** no benefit; ↑ SSI
- **Nichols–Condon 1977:** oral neomycin + erythromycin; sepsis 43% → 9%

- **Oral antibiotic prep drawback:** *C. difficile*
- **PEG** (polyethylene glycol) preferred: elderly, renal, CHF, electrolytes, cirrhosis — non-absorbable
- **Enema still used:** transrectal prostate biopsy
- **Descending colon flora:** 10^{10} – 10^{12} organisms/g
- **Ileus >72–96 h** → look for mechanical obstruction
- **Alvimopan** (Food and Drug Administration 2008) ↓ stay after cystectomy; gum chewing helps

INTRAOPERATIVE MANAGEMENT

- **Kurz 1996:** hypothermia → $3\times$ wound infection, +2.6 days
- **Mild hypothermia SSI range:** 34–36°C
- **Harvard Medical Practice Study 1991:** 48% adverse events surgical
- **Femoral neuropathy:** retractor blade on psoas
- **Neuropathy:** 2nd commonest anaesthetic liability claim
- **Arm abduction $\leq 90^\circ$** supine
- **Lithotomy:** both legs together; hips 80–100° flexion, 30–45° abduction
- **Flank:** axillary roll caudal to axilla
- **Nerve map:** ulnar groove; radial — spiral groove; median — elbow extension; sciatic — hamstring stretch; peroneal — fibular head
- **Hip position:** no effect on femoral neuropathy
- **Steep Trendelenburg:** avoid shoulder braces
- **SSI:** ~40% of nosocomial infections; +7–10 days
- **Hair:** clippers, immediately before
- **Timing:** ≤ 30 min better than 31–60; at/after incision → $\sim 2\times$ SSI
- **Wound classes:** clean / clean-contaminated (controlled entry) / contaminated (major break, transrectal ultrasound (TRUS) biopsy) / dirty
- **AUA** (American Urological Association) **default:** single dose; PCNL (percutaneous nephrolithotomy) and prosthetics ≤ 24 h
- **Cystoscopy + sterile urine:** no prophylaxis
- **Transurethral resection (prostate, bladder tumour):** cefazolin or trimethoprim-sulfamethoxazole
- **TRUS biopsy:** fluoroquinolone or cephalosporin + aminoglycoside
- **Colon:** add anaerobic cover (metronidazole)
- **Prosthetics:** aminoglycoside + cephalosporin or vancomycin; routine vancomycin not recommended
- **Aztreonam** replaces aminoglycoside in renal insufficiency
- **SWL** (shock-wave lithotripsy): only with risk factors; penile surgery — likely none
- **AUA compliance:** <60%

VTE PROPHYLAXIS

- **Urology VTE:** 10–40% unprophylaxed (historic); 1–5% now
- **VTE:** most important non-surgical death in urology
- **Prophylaxis:** ↑ minor, not major, bleeding
- **Low risk** (<40 y, minor, no factors): early ambulation
- **Moderate/high:** UFH 5000 U 12-/8-hourly, or enoxaparin 40 mg daily, or intermittent pneumatic compression (IPC) if bleeding risk
- **Highest:** drug + IPC
- **Enoxaparin** 30 mg if creatinine clearance <30 mL/min
- **LMWH:** less heparin-induced thrombocytopenia, irreversible; **UFH:** reversible, safe in renal failure

ANAESTHESIA, SKIN, TRANSFUSION, PAIN

- **Spinal hypotension** 10–40%; post-dural puncture headache <2% (29G pencil-point); serious deficit 0.05%

- **MAC** (minimum alveolar concentration): no movement to incision in 50%
- **Nitrous oxide:** contraindicated — bowel obstruction, pneumothorax
- **Halothane:** hepatitis 1 in 35,000; catecholamine sensitisation
- **Sevoflurane:** inhalational induction, difficult airway, bronchospasm
- **Desflurane:** faster recovery >3 h
- **Ketamine:** hypovolaemic, asthmatic induction
- **Propofol:** bronchodilator, nausea-free; hypotension
- **Succinylcholine:** only depolariser; ~5 min; malignant hyperthermia, hyperkalaemia
- **TOF (train-of-four)** ≥ 0.9 (was 0.7); quantify before extubation
- **Chlorhexidine 0.5% in spirit** edges alcoholic povidone-iodine (clean surgery)
- **Packed red cell Hct** ~70% vs whole blood 40%
- **Platelets:** <50,000 + surgery; 50–100,000 + high-risk/dysfunction
- **Fresh frozen plasma:** warfarin reversal, factor deficit, bleeding + INR >1.5; after 1 blood volume
- **Viral risk:** human immunodeficiency virus/hepatitis C ~1 in 2 million; hepatitis B ~1 in 200,000
- **Platelet bacterial contamination:** ~1 in 5000
- **Massive transfusion protocol:** anticipated >10 units/24 h
- **New persistent opioid use:** ~6% after surgery
- **NSAIDs:** avoid in renal insufficiency
- **Epidural + general anaesthesia:** ↓ perioperative mortality

CLASSIC TRAPS

- **Trap:** stop smoking the week before — **Actually:** <8 weeks may ↑ risk
- **Trap:** wait 3–6 months after MI — **Actually:** 4–6 weeks
- **Trap:** modified index uses Goldman weights — **Actually:** 1 point each
- **Trap:** thyroid storm in hypothyroid — **Actually:** hyperthyroid
- **Trap:** MBP reduces leak — **Actually:** no benefit, ↑ SSI
- **Trap:** antibiotic at incision fine — **Actually:** at/after incision doubles SSI
- **Trap:** femoral neuropathy from hip flexion — **Actually:** psoas retractor compression
- **Trap:** axillary roll in axilla — **Actually:** just caudal
- **Trap:** TPN helps all malnourished — **Actually:** harms well/mildly malnourished
- **Trap:** NSAIDs safe in pregnancy — **Actually:** ductus closure; use opioids